NACHAPPA PP

Smart manufacturing, Industry 4.0 systems, AI-driven industrial analytics, and Digital Transformation & Operational Excellence.

Email: npp.nachappa@gmail.com
Ph: +91-9108330616 | LinkedIn: NACHAPPA PP | Engineering Portfolio: View Portfolio
Bengaluru, Karnataka, India

Skills

Technologies & Tools: Python | Machine Learning | Computer Vision | Industry 4.0 | Industrial IoT (IIoT) | Smart Manufacturing | Manufacturing Analytics | Digital Transformation | OPC-UA | MES | OEE Analytics | Embedded Systems
Languages: C | C++ | Java | JavaScript | TypeScript | X++ | C# | Go | Rust | SQL | Kotlin | Swift | PHP | R | MATLAB | VHDL | Verilog | Shell Scripting
Automotive & Embedded Systems: CAN Bus | UDS | ECU Communication | Automotive Diagnostics | Embedded Systems | MATLAB | Simulink

Work Experience

ACE DESIGNERS LTD., Bengaluru
Engineer – Operational Excellence & Digital Transformation | Mar 2025 – Present

- Led process improvement through operational excellence along with 10+ digital transformation initiatives across Assembly, Design, Supply Chain, Quality, and Production Planning functions, improving production visibility, decision-making, operational efficiency by 10–20% in a year, and delivering supply chain cost savings of ₹25–30 Lakhs through strategy to utilise aged inventory in FY2025-26.
- Conceptualized, developed, and deployed the "Plant on Palm" manufacturing intelligence platform, currently used plant-wide and by 5,000+ users, providing real-time visibility into assembly progress, dispatch readiness, trials, and customer handovers, reducing customer status enquiries by 40–50%.
- Digitized end-to-end quality inspection workflows across CNC flowline and assembly operations, eliminating paper-based documentation and achieving 100% digital traceability across inspection and validation processes.
- Established weekly Theory of Constraints (TOC) review frameworks and operational governance mechanisms, improving bottleneck identification, throughput optimization, and cross-functional execution across manufacturing operations.
- Improved delivery performance from 29% to 45%, increased on-time delivery commitment accuracy by 45%, and reduced inspection cycle time by 2–3 days through process improvements, digitalization, and cross-functional collaboration.

Central Manufacturing Technology Institute, Bengaluru
Project Associate – II (Smart Manufacturing & Industry 4.0) | Jul 2023 – Feb 2025

- Led the development and deployment of a Quality Management Suite (QMS) solution for aerospace and precision manufacturing organizations, improving digital inspection, process traceability, and quality visibility while reducing manual reporting efforts by 80%.
- Developed Industry 4.0 solution deployment and smart manufacturing assessments at Hindustan Aeronautics Limited (HAL), Bharat Electronics Limited (BEL), and HAL Technical Training Institute (HAL-TTI), enabling digital manufacturing readiness in Govt. PSU.
- Planned and executed Manufacturing Execution System (MES) implementation for the Fabrication Components Division at BEL, improving production tracking and real-time shopfloor visibility
- Developed a smart tool holder solution for aerospace manufacturing applications, enabling real-time cutting parameter monitoring, tool utilization tracking, and predictive maintenance capabilities, contributing to 30% reduction in tooling-related downtime.

Education

ACHARYA INSTITUTES, Bengaluru
B.E. in Electronics & Communication Engineering

Aug 2019 – May 2023
CGPA: 8.55/10

Relevant Coursework: Embedded Systems and Microcontrollers, Digital Signal Processing, Industrial IoT, Computer Networks, fundamentals of AI, Machine Learning and Data Structures & Algorithms, Control Systems, VLSI Design, Communication Systems, Industrial Automation.

Projects

AI-Based Smart Material Allocation System (2025): Developed a machine learning-based inventory allocation system to optimize material planning, reducing shortages by 35% and improving procurement efficiency by 12%. Python, Machine Learning, Manufacturing Analytics.

- Predictive Vehicle Maintenance using Machine Learning (Research Project): Developed a predictive analytics model to identify potential vehicle component failures using sensor and operational data. Machine Learning, Python, Predictive Analytics.
- CAN Bus-Based Vehicle Diagnostics Platform (Academic Project): Designed a vehicle diagnostics solution capable of reading and analysing ECU parameters through CAN communication for fault detection and health monitoring. CAN Bus, Automotive Diagnostics, Embedded Systems.
- Battery Health Monitoring System for Electric Vehicles (Concept Project): Designed an AI-assisted battery monitoring framework for estimating State of Charge (SOC) and predicting battery degradation trends in EV applications. Machine Learning, Embedded Systems, EV Technologies.

Awards & Certificates

- Unique Contributor Award FY2025-26 | ACE DESIGNERS LTD., Bengaluru
- Best Technology Trainer FY2025-26 | ACE DESIGNERS LTD., Bengaluru
- Best Outgoing Student (B.E – Electronics and Communication) | Acharya Institutes., Bengaluru